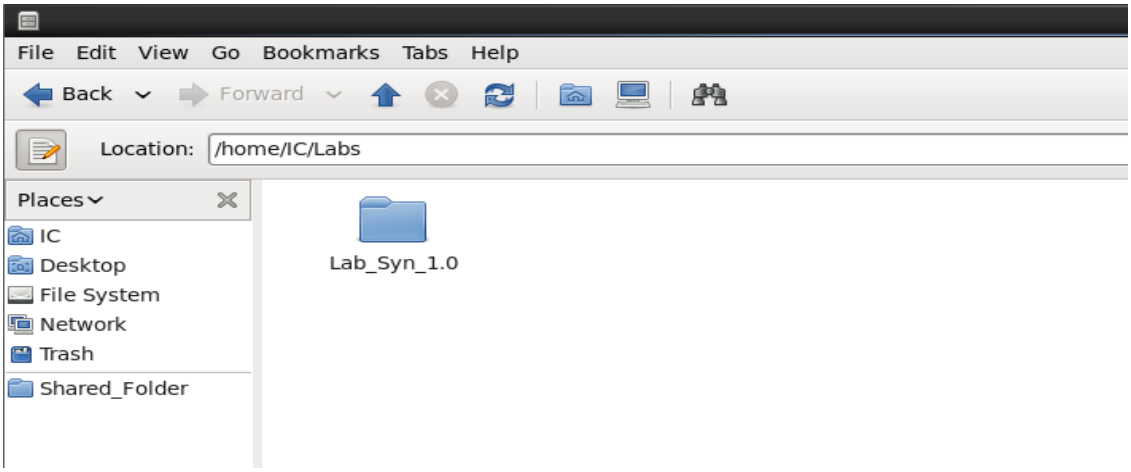
Lab_Syn_1.0

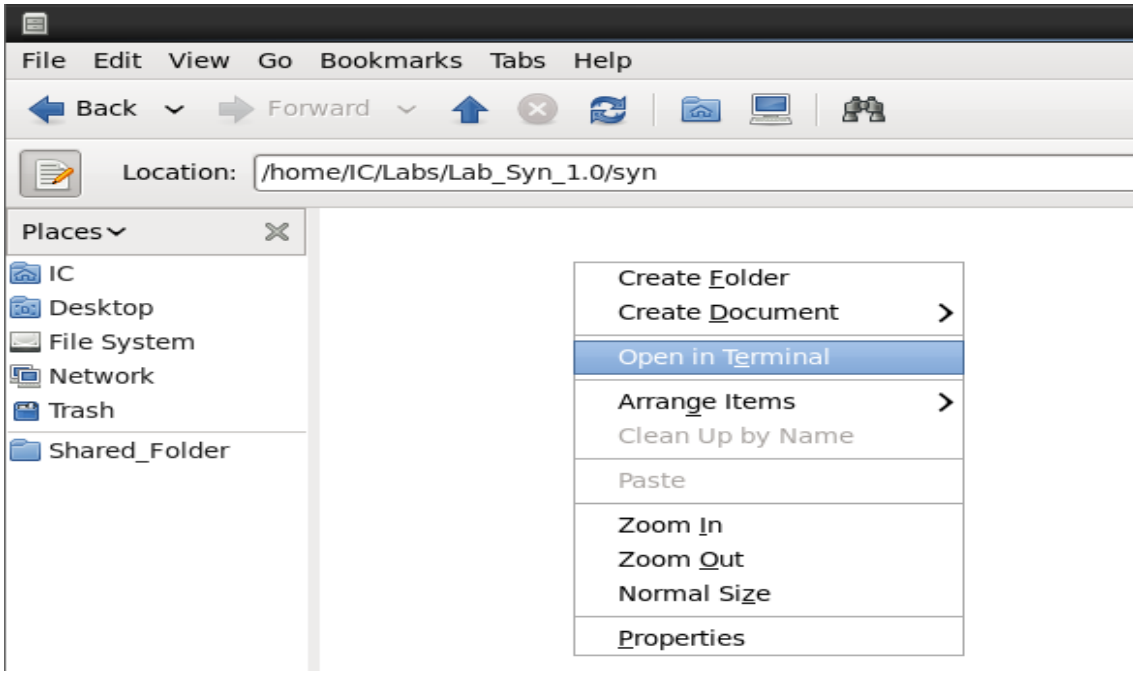
You have to synthesize Up_Dn_Counter.v file using Design Compiler GUI and generate Technology Dependent gate level netlists in Verilog format using standard cell library (scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db)

Steps: -

1. Copy Lab_Syn_1.0 folder from your computer into the virtual machine inside Labs directory



2. Open the terminal inside path /home/IC/Labs/Lab_Syn_1.0/syn



3. Open design Compiler tool in GUI Mode by run design_vision command in the terminal

4. Load standard cell Library in db format
   1) Add standard cell libraries to target_library
   2) Add standard libraries to link_library

5. Reading Up_Dn_Counter.v rtl File Using Read Command

6. Link the design

Once you are reading a RTL file, design compiler will do the followings: -

- Load GTECH library (To use it during translation of rtl to generic gates)
- Load the standard cell Libraries (To use in the mapping step)
- Checking HDL language syntax of the rtl files using PRESTO Compiler

```
design_vision> read_file -format verilog {/home/IC/Labs/Lab1.0/rtl/Up_Dn_Counter.v}
Loading db file '/home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db'
Loading db file '/home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db'
Loading db file '/home/IC/tsmc_fb_cl013g_sc/aci/sc-m/synopsys/scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.db'
Loading db file '/opt/Synopsys/Synplify2015/libraries/syn/gtech.db'
Loading db file '/opt/Synopsys/Synplify2015/libraries/syn/standard.sldb'
  Loading link library 'scmetro_tsmc_cl013g_rvt_tt_1p2v_25c'
  Loading link library 'scmetro_tsmc_cl013g_rvt_ss_1p08v_125c'
  Loading link library 'scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c'
  Loading link library 'gtech'
Loading verilog file '/home/IC/Labs/Lab1.0/rtl/Up_Dn_Counter.v'
Detecting input file type automatically (-rtl or -netlist).
Reading with Presto HDL Compiler (equivalent to -rtl option).
Running PRESTO HDLC
Compiling source file /home/IC/Labs/Lab1.0/rtl/Up_Dn_Counter.v

Inferred memory devices in process
        in routine Up_Dn_Counter line 20 in file
                '/home/IC/Labs/Lab1.0/rtl/Up_Dn_Counter.v'.
===============================================================================
|     Register Name     |    Type    | Width | Bus | MB | AR | AS | SR | SS | ST |
===============================================================================
|      Counter_reg      | Flip-flop  |   5   |  Y  | N  | N  | N  | N  | N  | N  |
===============================================================================
Presto compilation completed successfully.
Current design is now '/home/IC/Labs/Lab1.0/rtl/Up_Dn_Counter.db:Up_Dn_Counter'
Loaded 1 design.
Current design is 'Up_Dn_Counter'.
design_vision>
Current design is 'Up_Dn_Counter'.
Log   History
```

7. View design schematic in gtech cells

8. Compile Up_Dn_Counter.v File Using Compile Command

9. View design schematic and find the inferred library cell name of Counter_reg[2] Flip Flop, its inputs and outputs

10. Generate the gate level netlist in Verilog format.

Repeat the Lab using Analyze/Elaborate instead of Read command